Causal analysis of ecology, subsistence composition, and social organization in trans-historical small-scale farming societies using mixed-type Bayesian networks

Rosy Oh[1], Man-Suk Oh[2*]

1 Department of Mathematics, Korea Military Academy, Seoul, Korea

2 Department of Statistics, Ewha Womans University, Seoul, Korea

* msoh@ewha.ac.kr (MO)

Abstract

Identifying how ecological conditions, subsistence strategies, and social organization are interrelated is a central question in anthropology and archaeology. We analyzed 157 trans-historical, cross-cultural small-scale farming societies from the D-PLACE and eHRAF databases with a causal Bayesian network learned by the PC algorithm, aiming to identify the direct and indirect pathways among ecological, subsistence, and social organization variables and to quantify the interventional effects of upstream variables on each subsistence and social organization variable. Continuous and categorical variables were analyzed jointly without discretization, subsistence shares were represented as log-ratios relative to agriculture, the dominant activity, to respect their compositional nature, and the direction of dependence between subsistence and social organization was learned from the data rather than prespecified. Network stability was assessed by bootstrap resampling. Higher precipitation reduced the relative importance of animal husbandry and gathering compared with agriculture, and higher elevation reduced that of fishing. Steeper terrain more than halved the probability of large extended households and increased that of nuclear households, and elevation had similar but weaker effects transmitted through slope. Greater distance from the coast increased the probability of

exogamous clan organization. Subsistence composition and social organization were also linked: smaller communities were associated with a higher relative importance of hunting and gathering, and a higher relative importance of gathering increased the probability of hamlet rather than village settlement. When all continuous variables were dichotomized, the links between gathering and social organization were retained but arcs involving hunting, fishing, and community organization were lost. These results suggest that different dimensions of social organization follow distinct ecological and economic pathways, and that retaining continuous information can be important in causal network analysis of cross-cultural data.

Introduction

Small-scale farming societies are embedded in socio-ecological systems in which environmental conditions, economic activities, and social institutions interact [1]. A long tradition in anthropology and archaeology has asked how far ecological conditions shape the ways such societies obtain food and organize themselves, and how much room remains for social choice and historical contingency. Addressing this question requires identifying how ecological conditions, subsistence strategies, and social organization are interrelated: which relationships are direct, which are transmitted through other variables, and how strong they are. Because direct evidence on social organization is scarce in the archaeological record, cross-cultural ethnographic data have been widely used to identify regularities among environment, subsistence, and social forms that can inform hypotheses about past and present societies [2, 3].

Several specific relationships have been proposed. Cross-cultural studies have linked household extension to the labor demands and organization of the local economy [4–6], suggesting that terrain conditions that fragment cultivable land may affect household form. Exogamous descent groups create alliances among local groups through marriage exchange [7], and such ties can buffer subsistence risk through reciprocity [8, 9], a function that may matter less where abundant marine resources are accessible [10]. In mixed farming economies, the relative importance of animal husbandry has been associated with upland environments and seasonal vertical mobility [11, 12], and greater reliance on wild resources is commonly associated with smaller and more mobile

residential groups [13]. These relationships rarely operate in isolation: ecological variables are themselves interrelated, and their effects on social organization may be transmitted through subsistence, or vice versa, along multiple direct and indirect pathways. Network-based approaches that represent the joint dependence structure among all variables are therefore well suited to identifying these pathways.

Bayesian networks (BNs), which represent conditional dependence relationships among variables through a directed acyclic graph (DAG) [14], have recently been applied to cross-cultural anthropological data. Hamilton et al. [15] used a BN constructed from discrete variables to show how institutional complexity in small-scale societies emerges from socioecological complexity. Most directly relevant to the present study, Palacios et al. [2] compiled 173 trans-historical, cross-cultural small-scale farming societies from the D-PLACE [16] and eHRAF [17] databases and analyzed them with BNs, asking whether ecological features and social organization constrain subsistence strategies, and whether ecological features and subsistence strategies constrain social organization. They used predictive BN classifiers built on discretized variables. In the naive Bayes and augmented naive Bayes classifiers, each subsistence or social organization variable was designated as the outcome and the remaining variables as inputs. The diagnosis-type networks, whose structures were learned by score-based hill-climbing, were constructed separately for ecology and subsistence and for ecology and social organization, so no single network included all three domains. They found that ecological conditions were associated with several subsistence activities, particularly hunting, fishing, and animal husbandry, and with community size, community organization, and household organization, but concluded that, apart from gathering, the relative proportions of subsistence activities were largely independent of social organization. They also noted that hybrid BNs accommodating both numerical and categorical variables might reveal relationships that their discretized models could not capture.

Features of the data and the research aims motivate a reanalysis. First, the ecological variables, community size, and subsistence shares are quantitative, whereas settlement type, community organization, and household organization are categorical. Discretizing the quantitative variables loses information and may obscure conditional dependencies. Mixed-type structure learning methods avoid this by accommodating continuous and categorical variables jointly. Second, subsistence shares sum to a constant and thus form

compositional data; analyzing the shares separately can induce spurious associations among them and with other variables, whereas log-ratio transformations represent their relative information in a principled way. Third, whether subsistence shapes social organization or social organization shapes subsistence is not established a priori, and relationships among the three domains may be transmitted along indirect pathways. In predictive classifiers the direction is fixed in advance by the choice of outcome, and networks learned separately for pairs of domains cannot represent pathways linking subsistence and social organization, or ecological effects transmitted through one of these domains to the other. Learning one causal network from all variables jointly allows the direction of relationships between subsistence and social organization to be estimated from the data, while still incorporating well-established constraints such as the upstream position of ecological variables. Fourth, the substantive interest lies in how outcomes would change if ecological, subsistence, or social conditions differed, which is an interventional rather than a predictive question. Under appropriate causal assumptions, a BN learned from data can be combined with Pearl's do-calculus [18] to estimate such interventional effects while accounting for indirect pathways.

In this study, we reanalyze 157 societies from the dataset of Palacios et al. [2] with two aims: to identify the direct and indirect pathways among ecological, subsistence, and social organization variables, and to quantify the interventional effects of upstream variables on each subsistence and social organization variable. We learn a causal BN from mixed continuous and categorical data using the constraint-based PC algorithm [19, 20] with conditional independence tests appropriate for mixed variable types, representing subsistence composition by log-ratios relative to agriculture, the dominant activity in the sample. Only basic structural constraints are imposed, reflecting well-established relationships among environmental variables and the compositional nature of the subsistence variables, while the direction of relationships between subsistence and social organization is learned from the data. We assess the stability of the network by bootstrap resampling, estimate the effects of hypothetical interventions on upstream variables by forward simulation, and evaluate the sensitivity of these effects to arcs whose direction could not be determined. Finally, we compare the results with those reported by Palacios et al. and with a discretized analysis of the same data, to assess which relationships are robust to the analytical approach and how

discretization affects the conclusions.

By identifying the pathways through which ecological conditions, subsistence composition, and social organization are linked and quantifying the effects transmitted along them, this study contributes to understanding socio-ecological relationships in trans-historical small-scale farming societies and illustrates how causal network analysis of mixed-type data can sharpen inference from cross-cultural records.

Materials and methods

Data source

This study uses variables from the cross-cultural dataset compiled by Palacios et al. [2], who systematically collected and curated ethnographic data from D-PLACE and the Human Relations Area Files (eHRAF) [16, 17, 21]. Their dataset comprises 173 ethnographic cases representing small-scale societies (fewer than 1,000 inhabitants) in which farming (agriculture and animal husbandry) constitutes more than 50% of subsistence, supplemented by hunting, gathering, and fishing; data quality was ensured through cross-validation between the two repositories, with inconsistent cases excluded.

Ecological values in the compiled dataset of [2] were cross-checked against the D-PLACE database [16]. Entries that were missing or differed from the D-PLACE values, the latter mostly apparent transcription errors such as an elevation value entered in the slope field and misplaced decimal points, were replaced by the D-PLACE values (S1 Table). These discrepancies did not alter the categories used in the discretized analysis of Palacios et al. [2] and therefore do not affect their conclusions.

Of the 173 societies, 16 were excluded: 14 had missing values in a social organization variable (community organization, 11; household organization, 2; settlement type, 1), and 2 had subsistence entries that did not correspond to a valid interval code. The final analytical sample therefore comprised 157 societies, whose geographic distribution is shown in Fig 1. The study used only published, publicly available ethnographic data and involved no human participants or archaeological specimens; no permits were required, and the study complied with all relevant regulations.

Variables

We selected variables organized into three conceptual domains: ecological (environmental) characteristics, subsistence strategies, and social organization (Table 1). Ecological variables originally included distance to the nearest coastline (DistanceCoast), mean elevation at the society location (Elevation), mean terrain slope (Slope), mean annual temperature (AnnualTemp), mean monthly precipitation (MonthlyPrecip), mean monthly net primary production (MonthlyPrimaryprod), and the corresponding coefficients of variation (cv.AnnualTemp, cv.MonthlyPrecip, cv.MonthlyPrimaryprod). We excluded the three coefficient of variation variables because their high correlations with the corresponding mean values would introduce multicollinearity in parameter estimation and, more critically for our purposes, create redundant conditional dependence signals that can obscure and distort causal structure learning.

Table 1. Summary of study variables across three domains. Continuous variables are summarized as mean (SD) on their original scales, and categorical variables (marked with *) as counts (%). [a]Includes 15 societies originally coded as Camp.

Variable	Description	Mean (SD) / *N* (%)
Environmental characteristics		
DistanceCoast	Distance to nearest coastline (km)	447.2 (450.2)
Elevation	Mean elevation at society location (m)	647.2 (698.1)
Slope	Mean terrain slope (degrees)	2.2 (2.7)
AnnualTemp	Mean annual temperature (°C)	21.0 (7.1)
MonthlyPrecip	Mean monthly precipitation (mm/month)	145.4 (78.7)
Subsistence strategies		
Hunt_Agri	Log-ratio of hunting to agriculture	−1.79 (1.17)
Gather_Agri	Log-ratio of gathering to agriculture	−2.28 (1.09)
Animal_Agri	Log-ratio of animal husbandry to agriculture	−1.48 (1.35)
Fish_Agri	Log-ratio of fishing to agriculture	−1.78 (1.17)
Social organization		
CommSize	Mean community size (number of people)	218.47 (145.89)
SettlementType*	Settlement type	
	Village	79 (50.3%)
	Homesteads	35 (22.3%)
	Hamlet[a]	43 (27.4%)
CommOrg*	Community organization	
	No exogamous clans	113 (72.0%)
	Clans	44 (28.0%)
HouseholdOrg*	Household organization	
	Nuclear	42 (26.8%)
	Small extended	87 (55.4%)
	Large extended	28 (17.8%)

Monthly primary productivity was also excluded because it represents an intermediate measure of ecological productivity that is strongly influenced by climatic conditions, particularly precipitation, while also being closely related to subsistence activities. To reduce conceptual overlap between the ecological and subsistence domains and to facilitate interpretation of the relationships between basic environmental conditions and subsistence strategies, the ecological domain was restricted to more fundamental environmental variables: distance from the coast, elevation, slope, annual temperature, and monthly precipitation.

Subsistence variables comprise the percentage contributions to total subsistence from hunting (Hunting), gathering (Gathering), animal husbandry (AnimalHusbandry), fishing (Fishing), and agriculture (Agriculture). In the original dataset, these variables were coded as ordered percentage intervals (0–5%, 6–15%, 16–25%, ..., 76–85%, and 86–100%); we converted each interval to a continuous score using its midpoint. Because these relative contributions constitute compositional data, they were transformed using an additive log-ratio (ALR) transformation with Agriculture, the dominant subsistence activity among the sampled societies, as the reference component: Hunt_Agri = log(Hunting/Agriculture), Gather_Agri = log(Gathering/Agriculture), Animal_Agri = log(AnimalHusbandry/Agriculture), and Fish_Agri = log(Fishing/Agriculture). Agriculture was therefore omitted as a separate variable after transformation.

Social organization variables include mean community size (CommSize), settlement type (SettlementType), community organization (CommOrg), and household organization (HouseholdOrg). CommSize was originally provided as five ordered interval categories (<50, 50–99, 100–199, 200–399, and >400 persons); we converted each bounded interval to a continuous score using its midpoint and assigned representative values of 25 and 450 to the open-ended categories <50 and >400, respectively. The three categorical variables were retained in their discrete form. SettlementType originally had four levels (Village, Homesteads, Hamlet, and Camp); because the Camp category contained only 15 societies, it was combined with Hamlet to avoid a sparse category. CommOrg records the presence or absence of exogamous clans as a marker of social organizational complexity [22]. HouseholdOrg comprises three household forms: Nuclear, Small extended, and Large extended.

Prior to analysis, all continuous variables (the five ecological variables, CommSize,

and the four ALR-transformed subsistence variables) were subjected to a Yeo–Johnson transformation using the `bestNormalize` R package [23] and subsequently standardized to mean 0 and standard deviation 1. The Yeo–Johnson transformation reduces skewness and improves approximate normality while accommodating zero or negative values, and standardization places all continuous variables on a common scale, facilitating stable conditional independence testing and parameter estimation.

Causal Bayesian network structure learning

Bayesian networks

A BN encodes the joint probability distribution of a set of variables through a DAG, in which nodes correspond to variables and directed edges represent conditional dependence structure. Each edge points from a parent node toward a child node, signifying that the child's distribution is directly conditioned on its parents. Under the Markov property, the joint distribution factorizes into a product of local conditional distributions [14, 24–26]:

$$P(X_1, \ldots, X_p) = \prod_{i=1}^{p} P(X_i \mid \Pi_i)\,, \tag{1}$$

where $\{X_1, \ldots, X_p\}$ denotes the full variable set and Π_i denotes the parent set of X_i.

Structural constraints

Domain knowledge was incorporated through a blacklist specifying arcs that are not permitted (Table 2). We imposed a hierarchical ordering within the ecological domain to reflect the assumed relationships among geographical features, topography, and climate. DistanceCoast and Elevation were treated as higher-level geographical characteristics, followed by Slope as a topographical variable, and AnnualTemp and MonthlyPrecip as climatic variables; arcs that violated this ordering were prohibited. The arc between DistanceCoast and Elevation was also prohibited. Both variables describe the geographical location of a society, and neither can be regarded as a cause of the other; their empirical association (inland locations tend to lie at higher elevations) reflects the geography of the sampled locations rather than a causal process. A directed

arc between them would imply, for example, that an intervention on elevation changes the distance to the coast, and would transmit such spurious effects through the network. Prohibiting this arc keeps both variables as exogenous geographical characteristics.

Table 2. Baseline blacklist constraints used in Bayesian network structure learning.

Domain	Variables	Prohibited arcs
Ecology		
(a) Geography	DistanceCoast, Elevation	Between (a)
(b) Topography	Slope	(b) → (a)
(c) Climate	AnnualTemp, MonthlyPrecip	(c) → {(a), (b)}
(d) Subsistence strategy	Animal_Agri, Hunt_Agri, Fish_Agri, Gather_Agri	(d) → {(a), (b), (c)}; among (d)
(e) Social organization	CommSize, HouseholdOrg, CommOrg, SettlementType	(e) → {(a), (b), (c)}

Arcs from subsistence strategy and social organization variables to ecological variables were prohibited, restricting ecological variables to upstream positions in the network. Arcs among the subsistence strategy variables were also prohibited. Because all four variables are log-ratios with agriculture as the common denominator, a change in the share of agriculture alters all of them simultaneously, so they are mutually dependent by construction rather than through causal processes. An arc such as Gather_Agri → Hunt_Agri would imply that intervening on the relative importance of gathering changes that of hunting, an interpretation that is not meaningful for compositional data. Because the directional relationship between subsistence strategies and social organization is not unequivocally established, arcs between these two domains were allowed in either direction and were learned from the data.

Structure learning with the PC algorithm

Given these constraints, we applied the constraint-based PC algorithm [19, 20] to learn the causal structure. The PC algorithm first identifies pairs of variables that are marginally dependent and then iteratively tests for conditional independence given progressively larger conditioning sets, removing edges where independence is found. Remaining edges are oriented by identifying v-structures and propagating orientation rules that avoid new v-structures and directed cycles. The result is a completed partially directed acyclic graph (CPDAG) representing the Markov equivalence class of

DAGs consistent with the observed conditional independence structure [19, 27, 28].

Structure learning relies on the causal Markov condition and faithfulness, under which the conditional independencies in the data correspond exactly to the separation properties of the true causal graph [19]. Valid causal interpretation of the learned network additionally requires three identification assumptions [29]: (1) causal sufficiency, that is, all common causes of the measured variables are included; (2) consistency, that is, the observed value of each variable equals its potential outcome under the corresponding intervention; and (3) positivity, that is, all exposure levels occur with positive probability in the population.

Structure learning was performed with the `tpc()` function of the R package `tpc` (version 1.0) [30], an implementation of the PC algorithm built on `pcalg` [31]. The domain-based constraints in Table 2 were supplied as a matrix of forbidden directed edges; the tier ordering option of `tpc()` was not used. To accommodate mixed continuous and categorical variables without discretization, conditional independence was assessed with the likelihood-ratio test under a conditional Gaussian distribution, implemented as `mixCItest()` in the R package `micd` [32, 33], at significance level $\alpha = 0.1$. The skeleton was estimated with the order-independent (stable) variant of the PC algorithm, and v-structures were oriented using the majority rule [28].

Bootstrap stability and consensus network

To assess the stability of the learned network, we employed bootstrap-based structure learning [26, 34]. We generated 10,000 bootstrap samples, each comprising 80% of the original observations drawn without replacement, and learned a network from each sample using the PC algorithm. Bootstrap networks that could not be converted to valid DAGs were excluded; this affected fewer than 30 of the 10,000 replicates (<0.3%) in our case. For each pair of variables, arc strength was calculated as the proportion of retained bootstrap networks containing an arc between them, regardless of direction. Direction probability was calculated as the proportion of arc occurrences supporting a given orientation, with undirected arcs contributing equally to both directions. Arcs with strength at least equal to a prespecified threshold and direction probability greater than 0.5 were selected. A consensus DAG was then constructed by sequentially adding the selected directed arcs, excluding any arc that would introduce a directed cycle, and

isolated nodes were removed.

Quantifying interventional effects

To quantify the effects implied by the learned network, we conducted an interventional analysis based on Pearl's do-calculus [18, 35]. Rather than asking how an outcome differs when an explanatory variable is merely observed at different values, we asked how the outcome would change if that variable were hypothetically set to a particular value. This distinction is important because an observed association may reflect the influence of other variables, whereas an intervention isolates the effect transmitted through the causal network [36]. For each variable that was an ancestor of a given outcome in the learned network, we estimated the probability distribution of the outcome under hypothetical interventions on that variable. These interventional probabilities were then used to quantify how changes in upstream variables would propagate through the network and affect downstream outcomes.

For the interventional analysis, continuous variables were dichotomized into Low and High levels at their medians, whereas categorical variables retained their original categories. For each ancestor variable W and outcome X, we estimated the probability of a specified outcome level under hypothetical interventions on W,

$$P(X = x \mid do(W = w)). \tag{2}$$

For continuous intervention variables, the effect was summarized by the ratio of interventional probabilities under High and Low interventions,

$$\text{Ratio} = \frac{P(X = x \mid do(W = \text{High}))}{P(X = x \mid do(W = \text{Low}))}, \tag{3}$$

where x denotes the High level for continuous outcomes or a specified category for categorical outcomes. A ratio greater than 1 indicates an increased probability of the specified outcome under the High intervention relative to the Low intervention, whereas a ratio below 1 indicates a decreased probability. For categorical intervention variables, we estimated and compared $P(X = x \mid do(W = w))$ separately for each category w.

Estimation by forward simulation

Interventional probabilities were estimated by stochastic forward simulation from the fitted BN. Given the consensus network, the conditional distribution of each node given its parents was estimated from the 157 societies: continuous nodes were modeled by linear Gaussian regression on their parents, with categorical parents entered as indicator variables, and categorical nodes by multinomial logistic regression. Nodes without parents were modeled by a normal distribution with the sample mean and standard deviation (continuous) or by the observed category frequencies (categorical).

To simulate under an intervention on W, values of W were generated independently of its parents. For a continuous W, $do(W = \text{Low})$ and $do(W = \text{High})$ were implemented by resampling with replacement from the observed values of W at or below its median and above its median, respectively, so that the intervened values retained the empirical distribution within each half. For a categorical W, all simulated observations were set to the specified category. The remaining variables were then generated in topological order from their fitted conditional distributions, so that the intervention affected only the descendants of W. For each intervention, 1,000,000 observations were simulated, and the interventional probability of an outcome level was estimated as its relative frequency in the simulated data. For example, the probability of a large community was estimated as the proportion of simulated observations with CommSize above its sample median. Monte Carlo 95% confidence intervals were obtained using the Wilson score interval for the probabilities and the delta method on the log scale for the ratios.

Sensitivity to arc orientation

Arcs retained in the consensus network whose direction probability was below 0.6 were regarded as having uncertain directions. For such arcs, we constructed alternative networks covering every combination of their orientations, keeping all other arcs unchanged and retaining only those that were valid DAGs. Each network was fitted to the data, and interventional effects were re-estimated by forward simulation as described above. The fit of the alternative networks was compared using the Bayesian information criterion (BIC), $\text{BIC} = -2 \log L + k \log n$, where L is the maximized likelihood, k the number of free parameters, and n the sample size, so that lower values

indicate better fit. Because the network contains arcs from continuous to categorical variables, which the conditional Gaussian model does not permit, the likelihood was obtained by fitting each node on its parents in the network, using linear regression for continuous nodes and multinomial logistic regression for categorical nodes, and summing the node-wise log-likelihoods and parameter counts.

Comparison with discretized Bayesian networks

To assess sensitivity to discretization, we constructed fully discrete versions of the same dataset. All continuous variables, including the four ALR-transformed subsistence coordinates, were dichotomized using the sample median. Originally categorical variables retained their categories. We applied the same structure-learning procedure with the same baseline blacklist constraints, significance level ($\alpha = 0.1$), and bootstrap-based arc aggregation, using `disCItest` for discrete conditional independence testing. The resulting consensus networks were compared with the mixed-type network at the same arc-strength threshold of 0.3.

Software and reproducibility

All analyses were performed in R (version 4.6.0) using the packages `tpc` (version 1.0) [30], `pcalg` (version 2.7.12) [31], `micd` (version 1.1.2) [32], `bnlearn` (version 5.1) [26], and `bestNormalize` (version 1.9.2) [23]. The processed dataset of 157 societies, including the corrected ecological values (S1 Table), and all R code, including the main analysis script, helper functions, and instructions for reproducing all reported analyses, are available at [xxxxx].

Results

Learned causal network

As the arc strength threshold increased from 0.2 to 0.3 and 0.4, the number of arcs in the consensus network decreased from 23 to 16 and 10, respectively. We selected the intermediate threshold of 0.3, which balances interpretability against structural stability: it yields a parsimonious network composed of arcs recovered in at least 30% of the

bootstrap networks, appropriate for the modest sample size ($n = 157$), while retaining more of the supported relationships than the stricter threshold of 0.4.

The learned network is presented in Fig 2. Dashed arcs (Gather_Agri → CommSize and CommSize → Hunt_Agri) indicate directional uncertainty, with bootstrap direction probabilities of 0.51 and 0.56, respectively. Bootstrap arc strengths and direction probabilities for all arcs are given in S2 Table.

The network exhibits a complex pattern of direct and indirect relationships among ecological characteristics, subsistence strategies, and social organization. Note that the structural constraints only prohibited arcs that violated the assumed ordering and did not require any arc to be present; all arcs in the network, including those among ecological variables, were therefore determined by the data. For example, an arc from DistanceCoast to Slope was permitted but not retained. Within the ecological domain, DistanceCoast and Elevation occupy the most upstream positions. DistanceCoast has directed arcs to AnnualTemp, MonthlyPrecip, and CommOrg, suggesting that coastal proximity may shape both local climatic conditions and community organization. Elevation affects AnnualTemp, Slope, and Fish_Agri, consistent with the effects of altitude on temperature, terrain, and access to aquatic resources. AnnualTemp further affects Hunt_Agri and SettlementType, and MonthlyPrecip affects Gather_Agri, Animal_Agri, and SettlementType. Slope directly affects HouseholdOrg, indicating that topographic constraints may extend to the organization of domestic groups.

Ecological influences on subsistence strategies operate through both direct and indirect pathways. Gather_Agri and Animal_Agri are each directly influenced by MonthlyPrecip and therefore have MonthlyPrecip and DistanceCoast as ecological ancestors. Fish_Agri is directly influenced by Elevation only. Hunt_Agri is directly influenced by AnnualTemp and CommSize and thus has DistanceCoast, Elevation, and MonthlyPrecip as further ecological ancestors, with the effect of MonthlyPrecip transmitted through Gather_Agri and CommSize.

Social organizational variables are embedded within these environmental and subsistence pathways. SettlementType is the most downstream node in the network: it is directly influenced by MonthlyPrecip, AnnualTemp, Gather_Agri, and Animal_Agri, and indirectly by DistanceCoast and Elevation, suggesting that settlement form may represent an integrated response to multiple ecological and economic conditions.

HouseholdOrg is directly influenced by Slope, and thus indirectly by Elevation, but has no descendants; at the selected threshold, household organization was therefore not linked to subsistence or to the other dimensions of social organization. CommOrg is directly influenced by DistanceCoast only. CommSize is linked to Gather_Agri and Hunt_Agri, but the directions of both arcs were uncertain (direction probabilities 0.51 and 0.56); because these directions determine whether CommSize acts as a cause or a consequence of subsistence composition, we fitted the network under all four combinations of their orientations and compared the model fit (BIC) and resulting interventional effects (see Sensitivity to arc orientation).

Interventional effects

Based on the BN structure shown in Fig 2, we assessed the interventional effects of upstream variables on subsistence strategies and social organization (Figs 3 and 4). For each outcome variable, we identified its ancestors in the learned network and estimated how interventions on these variables altered the probability distribution of the outcome. The estimated interventional effects are presented as bar plots. Error bars representing 95% Monte Carlo confidence intervals were omitted because the large simulation size (1,000,000 observations per intervention) yielded intervals too narrow to be visually distinguishable in the plots.

Subsistence strategies

Fig 3 presents the interventional effects on subsistence strategy variables. Because the four subsistence strategy variables are log-ratios relative to agriculture, a "high" outcome denotes a relative contribution above the median compared with agriculture.

Animal_Agri. MonthlyPrecip and DistanceCoast were the only ancestors in the learned network. Higher MonthlyPrecip substantially decreased the probability of a high animal husbandry-to-agriculture ratio (ratio 0.70), whereas higher DistanceCoast increased it (1.15). These results suggest that wetter conditions favor agriculture relative to animal husbandry, and that the relative importance of animal husbandry tends to be greater in inland environments.

Hunt_Agri. Higher Gather_Agri, Elevation, and DistanceCoast increased the

probability of a high hunting-to-agriculture ratio, with probability ratios of 1.24, 1.22, and 1.12, respectively. In contrast, higher AnnualTemp and CommSize substantially reduced this probability, with ratios of 0.65 and 0.56, whereas the effect of MonthlyPrecip was small (0.96). Because Elevation and Gather_Agri act on Hunt_Agri through AnnualTemp and CommSize, respectively, hunting appears relatively more important in cooler environments and in smaller communities, and therefore also where gathering is prominent and at higher elevations.

Gather_Agri. MonthlyPrecip and DistanceCoast were the only ancestors. Higher MonthlyPrecip decreased the probability of a high gathering-to-agriculture ratio (ratio 0.80), suggesting that the relative importance of gathering declines under wetter conditions, potentially reflecting a greater relative contribution of agriculture. Higher DistanceCoast had a small positive effect (1.10).

Fish_Agri. Elevation was the only ancestor in the learned network. Higher Elevation substantially decreased the probability of a high fishing-to-agriculture ratio (ratio 0.56), suggesting that elevation constrains the relative importance of fishing, potentially through reduced access to aquatic resources.

Overall, precipitation had negative estimated effects on the relative importance of both animal husbandry and gathering, distance from the coast had positive effects on animal husbandry, gathering, and hunting, elevation had a particularly strong negative effect on fishing, and hunting was shaped by a broader combination of ecological, subsistence, and social organizational factors.

Social organization

Fig 4 presents the interventional effects on community size, household organization, community organization, and settlement type.

Community size. Higher Gather_Agri produced the strongest effect, substantially reducing the probability of a large community (ratio 0.60). Higher MonthlyPrecip increased this probability (1.11), whereas DistanceCoast had a small negative effect (0.96); both acted through Gather_Agri. Larger communities were thus less likely when gathering was relatively important compared with agriculture, and more likely under wetter conditions that reduced the relative importance of gathering.

Household organization. Slope and Elevation were the only ancestors, with

Elevation acting through Slope. Higher Slope more than halved the probability of large extended households (ratio 0.46) and increased the probability of nuclear households (1.38), with a small increase for small extended households (1.09). Higher Elevation had effects in the same direction but of smaller magnitude (0.61, 1.23, and 1.06 for large extended, nuclear, and small extended households, respectively). Steeper terrain therefore disfavored large extended households and favored nuclear households.

Community organization. DistanceCoast was the only ancestor. Greater distance from the coastline increased the probability of exogamous clan organization (ratio 1.67) and decreased the probability of having no exogamous clans (0.82).

Settlement type. Effects differed substantially across settlement categories. Higher MonthlyPrecip and AnnualTemp increased the probability of village settlement (ratios 1.30 and 1.12), whereas higher Gather_Agri substantially reduced it (0.68); DistanceCoast, Animal_Agri, and Elevation had small negative effects (0.88, 0.90, and 0.95). For homesteads, most effects were modest; higher AnnualTemp, Animal_Agri, and DistanceCoast had small positive effects (1.11, 1.10, and 1.06), and MonthlyPrecip had the strongest negative effect (0.82). Hamlet settlement showed a particularly strong positive response to Gather_Agri (2.19), followed by smaller positive effects of DistanceCoast (1.20), Elevation (1.15), and Animal_Agri (1.12), whereas higher AnnualTemp and MonthlyPrecip reduced its probability (0.74 and 0.73).

Sensitivity to arc orientation

Two arcs in Fig 2 had uncertain directions: the Gather_Agri→CommSize arc had high bootstrap strength (0.91) but a direction probability close to 0.5 (0.51), and the CommSize→Hunt_Agri arc (strength 0.36) had a direction probability of 0.56. We refer to the network in Fig 2, in which the directions of arcs between the subsistence and social organization domains were learned from the data, as M0. From M0, we constructed three alternative networks by reversing Gather_Agri → CommSize (M1), CommSize → Hunt_Agri (M2), or both arcs (M3), keeping all other arcs unchanged; all three alternative networks were valid DAGs. The interventional effects under M1, M2, and M3 are shown in S1–S2, S3–S4, and S5–S6 Figs, respectively.

The interventional effects on Animal_Agri, Fish_Agri, HouseholdOrg, and CommOrg

were identical in M0–M3 up to Monte Carlo error, because the ancestors of these outcomes and their conditional distributions are the same in all four networks. The same holds for the effects of Gather_Agri and Animal_Agri on SettlementType, whose parents are unchanged; higher Gather_Agri decreased the probability of village settlement (ratio 0.68) and substantially increased that of hamlet settlement (2.18–2.19) in all networks. Effects transmitted through the reversed arcs could differ among the networks but were stable in practice: the effects of AnnualTemp, Elevation, and DistanceCoast on Hunt_Agri (ratios 0.65–0.67, 1.19–1.22, and 1.09–1.12, respectively), the negative effect of MonthlyPrecip on Gather_Agri (0.80–0.82), and the effects of the ecological variables on SettlementType varied only slightly across M0–M3.

The effects that depended on orientation were those transmitted among Gather_Agri, CommSize, and Hunt_Agri. In M1, CommSize had no ancestors, so the negative effect of Gather_Agri on CommSize and the effects of the ecological variables acting through it disappeared. Conversely, higher CommSize reduced the probability of a high gathering-to-agriculture ratio (ratio 0.66), and CommSize became an ancestor of SettlementType (1.14, 1.03, and 0.77 for village, homesteads, and hamlet settlement, respectively). Gather_Agri and MonthlyPrecip were no longer ancestors of Hunt_Agri, whereas the negative effect of CommSize on Hunt_Agri was unchanged (0.56). In M2, CommSize and the variables acting through it were no longer ancestors of Hunt_Agri; Gather_Agri still reduced the probability of a large community, although less strongly than in M0 (0.67 vs. 0.60), because part of its association with CommSize was now attributed to Hunt_Agri, and higher Hunt_Agri reduced the probability of a large community (0.86). The ancestors of Hunt_Agri thereby also became ancestors of CommSize, with small effects of AnnualTemp, Elevation, and DistanceCoast (1.05, 0.98, and 0.96), whereas higher MonthlyPrecip still increased it through Gather_Agri (1.09 vs. 1.11 in M0). In M3, the effects on Hunt_Agri were as in M2, and CommSize again reduced Gather_Agri (0.66). Higher Hunt_Agri reduced the probability of a large community (0.74), and higher AnnualTemp increased it through Hunt_Agri (1.09), whereas the effects of DistanceCoast and Elevation remained small (0.98 and 0.96). Hunt_Agri also became an ancestor of Gather_Agri and SettlementType through CommSize, with small effects (ratios between 0.97 and 1.10).

Thus, the principal ecological effects on subsistence strategies and on household

organization, community organization, and settlement type, as well as the effects of gathering on settlement type, were robust across M0–M3, whereas the effects among Gather_Agri, CommSize, and Hunt_Agri, and the downstream effects of CommSize on SettlementType, depended on the assumed orientations.

The four networks differed in their fit to the data. M0 had the lowest BIC (5173.9), followed closely by M1 (5174.8; $\Delta\text{BIC} = 0.9$), whereas M3 (5179.0; $\Delta\text{BIC} = 5.1$) and M2 (5190.6; $\Delta\text{BIC} = 16.7$) fitted worse. The data therefore did not distinguish between the two orientations of the Gather_Agri–CommSize arc, whereas the networks with CommSize → Hunt_Agri fitted better than those with the reverse orientation. Taken together, the sensitivity analysis favors the orientation from community size to the relative importance of hunting, whereas the direction of the relationship between gathering and community size remains undetermined.

Comparison with discretized Bayesian networks

With median-based dichotomization of all continuous variables, the consensus network at the arc-strength threshold of 0.3 contained eight arcs (S3 Table and S7 Fig). Seven of these corresponded to arcs of the mixed-type network: the four arcs among ecological variables (DistanceCoast → AnnualTemp, DistanceCoast → MonthlyPrecip, Elevation → Slope, and Elevation → AnnualTemp), MonthlyPrecip → Animal_Agri, and the arcs linking Gather_Agri with CommSize and with SettlementType. The direction of the CommSize–Gather_Agri arc was again uncertain (direction probability 0.54), and the SettlementType–Gather_Agri arc was oriented from SettlementType to Gather_Agri (0.60), opposite to its direction in the mixed-type network. The remaining arc linked Elevation directly to HouseholdOrg, whereas in the mixed-type network this dependence was transmitted through Slope. Nine of the 16 arcs of the mixed-type network were not recovered: all arcs involving Hunt_Agri, Fish_Agri, and CommOrg, which were isolated in the discretized network, as well as MonthlyPrecip → Gather_Agri, Slope → HouseholdOrg, and the arcs from Animal_Agri, AnnualTemp, and MonthlyPrecip to SettlementType.

Discussion

This study examined how ecological conditions, subsistence composition, and social organization are interrelated in small-scale farming societies, by learning the direct and indirect pathways among them and quantifying the interventional effects transmitted along these pathways. Analyzing 157 trans-historical societies with a mixed-type causal BN, we found that ecological conditions affected subsistence composition and social organization through outcome-specific pathways, and that subsistence composition and social organization were not independent of each other.

Ecological effects on subsistence composition differed across activities. Higher precipitation reduced the relative importance of animal husbandry and, to a lesser extent, of gathering compared with agriculture, consistent with rainfall favoring cultivation over alternative activities. Inland location increased the relative importance of animal husbandry, gathering, and hunting, although these effects were modest in magnitude. Neither elevation nor slope was an ancestor of animal husbandry, so the proposed association of herding with upland environments and seasonal vertical mobility in mixed farming economies [11, 12] was not supported in this sample. Higher elevation markedly reduced the relative importance of fishing, plausibly reflecting reduced access to aquatic resources, whereas hunting responded to a broader set of conditions and was relatively more important in cooler settings, with the effect of elevation on hunting transmitted through temperature.

Links from social organization to subsistence composition were limited. Larger communities were associated with a lower relative importance of hunting, consistent with the depletion of game around larger settlements [37]. The bootstrap direction probability of this arc was only 0.56, but networks in which community size affected hunting fitted the data better than those with the reverse orientation, and its effect was unchanged when the direction of the gathering–community size arc was reversed. Household organization was not linked to any subsistence variable at the selected threshold, so a possible role of household labor in the balance between herding and cultivation [5, 6] remains to be established.

Ecological conditions also affected social organization, again through outcome-specific pathways. Household structure responded strongly to topography:

steeper terrain more than halved the probability of large extended households and favored nuclear households, and elevation had similar but weaker effects transmitted through slope. Cross-cultural studies have linked household extension to the labor demands and organization of the local economy [4–6]; rugged terrain, by fragmenting cultivable land and limiting the scale of coordinated work, may reduce the returns to pooling labor in large extended households. Community organization was most strongly affected by distance from the coast, with inland location increasing the probability of exogamous clan organization. Exogamous descent groups create alliances among local groups through marriage exchange [7], and such inter-group ties can buffer subsistence risk through reciprocity [8,9]; coastal societies, with access to abundant and diverse marine resources [10], may depend less on these institutions. Intergroup competition offers an alternative explanation for the development of descent-based groups [38].

The links between subsistence composition and social organization are of particular interest, because their directions were not prespecified but learned from the data. A higher gathering-to-agriculture ratio was associated with smaller communities and substantially increased the probability of hamlet rather than village settlement, consistent with the general association between reliance on wild resources and smaller, more dispersed residential groups [13]. The direction of the link between gathering and community size could not be determined from the data: its bootstrap direction probability was 0.51, and the two orientations fitted the data almost equally well. The effect of gathering on community size therefore depends on the assumed orientation, whereas the effects of gathering on settlement type were the same under all orientations. Together with the link between community size and hunting, these results indicate that subsistence composition and social organization are interrelated, and that different dimensions of social organization are associated with distinct ecological and subsistence pathways rather than with a common set of environmental factors.

Several of the relationships identified here correspond to those reported by Palacios et al. [2] for the same ethnographic sample, including the dependence of the relative importance of animal husbandry on precipitation, of community organization on distance to the coast, and of household organization on elevation (here transmitted through slope). The clearest difference concerns the link between subsistence and social organization. Palacios et al. concluded that, apart from gathering, the relative

proportions of subsistence activities were largely independent of social organization, and found no effect of social organization on hunting, animal husbandry, or fishing. The present network likewise links gathering with community size and settlement type, but it also contains a direct link from community size to hunting and from animal husbandry to settlement type, neither of which was reported previously. Some other relationships reported previously appeared here only as indirect pathways; for example, the dependence of hunting on coastal distance was transmitted through temperature, and that on precipitation through gathering and community size. Such mediated dependence cannot be distinguished from direct dependence in naive Bayes classifiers, which link each predictor directly to the outcome and were among the models used by Palacios et al. One notable discrepancy concerns fishing, which Palacios et al. related to coastal distance but which depended only on elevation here. Elevation and coastal distance are positively correlated in the sample ($r = 0.34, p < 0.001$), although the structural constraints prevented this association from being represented as an arc because neither can be regarded as a cause of the other. Given elevation, coastal distance was not retained as a parent of Fish_Agri, so the previously reported dependence of fishing on coastal distance may be partly accounted for by elevation, and the effect of elevation may reflect access to aquatic resources more broadly rather than an effect of altitude per se.

These comparisons are necessarily qualitative, because the two analyses differ in learning approach, sample, variable set, and representation of subsistence. To examine how discretization alone affects the conclusions, we repeated the analysis on the same societies after dichotomizing all continuous variables at their medians, keeping the structural constraints, significance level, bootstrap aggregation, and arc-strength threshold unchanged. The discretized network recovered the main ecological structure and the links of gathering with community size and settlement type, but lost all arcs involving hunting, fishing, and community organization, and replaced the dependence of household organization on slope with a direct dependence on elevation, as reported by Palacios et al. [2]. The pattern thus resembles their conclusion that, apart from gathering, subsistence composition was largely independent of social organization. Because the discrete and mixed-type analyses also differ in their conditional independence tests, and discretization produces sparse conditional contingency tables at

this sample size, this comparison indicates the practical consequence of discretization for the present data rather than isolating information loss as its sole cause. Nevertheless, it suggests that the apparent independence between subsistence composition and social organization in discretized analyses may partly reflect reduced power to detect dependence rather than its absence.

Several limitations should be acknowledged. First, causal interpretation depends on the validity of the learned network and on the assumptions described in Materials and methods, including faithfulness and the absence of important unmeasured confounders. Second, the modest sample size may affect the stability of the learned structure; we therefore based the network on bootstrap arc strengths, reported the strength and direction probability of every retained arc (S2 Table), and noted that the number of arcs varied with the choice of threshold, so that weakly supported relationships may have been omitted or included. Third, the directions of the arcs linking CommSize with Gather_Agri and Hunt_Agri were uncertain in the bootstrap analysis; model comparison favored CommSize $\rightarrow$ Hunt_Agri but did not resolve the direction of the Gather_Agri–CommSize arc, and the effects among these three variables depend on their orientations, whereas the principal ecological effects do not. Fourth, the structural constraints prohibited a direct arc between coastal distance and elevation, so the fitted network treats them as independent despite their positive correlation, and effects of coastal distance transmitted through elevation, such as on fishing, are not captured by the interventional analysis. Fifth, subsistence shares and community size were recorded as intervals and converted to midpoint scores, which introduces measurement error that may attenuate estimated dependencies. Sixth, because the sample was restricted to societies in which farming provides more than half of subsistence, the relative importance of agriculture, the reference component of all log-ratios, varies over a limited range; the subsistence variables should accordingly be interpreted as relative composition rather than absolute changes in individual activities. Seventh, the four subsistence log-ratios share agriculture as a common denominator and are therefore mutually dependent by construction, but arcs among them were prohibited because such arcs would have no causal interpretation. Their dependence that is not explained by the ecological variables could thus not be represented directly, and may instead have been captured by the links through community size, which provide the only path

between gathering and hunting in the network (Gather_Agri → CommSize → Hunt_Agri). Eighth, the interventional analysis considered single-variable interventions, whereas ecological and economic changes are likely to occur jointly. Finally, the sample comprises ethnographically documented societies, so caution is warranted when generalizing the findings to other socio-ecological contexts, including prehistoric ones.

Conclusions

Applied to 157 trans-historical small-scale farming societies, a mixed-type causal BN combined with a compositional representation of subsistence and interventional analysis showed that ecological conditions shape subsistence composition and distinct dimensions of social organization through outcome-specific pathways, with strong topographic effects on household organization and an effect of coastal distance on clan organization. Contrary to the conclusion of an earlier discretized analysis of these societies, subsistence composition and social organization were linked: a higher relative importance of gathering was associated with smaller communities and with hamlet rather than village settlement, and larger communities with a lower relative importance of hunting. When the same data were dichotomized, the links of gathering with community size and settlement type were recovered but that between community size and hunting was not, mirroring the earlier conclusion; the directions of the links involving community size, however, could not be fully resolved. The approach is applicable to other observational studies involving mixed variable types, hierarchical domain knowledge, and interconnected outcomes.

Data availability

The dataset analyzed in this study is derived from the publicly available cross-cultural ethnographic dataset of [2], available via PLOS ONE at `https://doi.org/10.1371/journal.pone.0276088`. The processed dataset of 157 societies, including the ecological values corrected against D-PLACE (S1 Table), and all R code used in this study are available at

https://github.com/mansukoh/BN-archaeology. The repository includes the processed dataset, the main analysis script, helper functions, and step-by-step instructions for reproducing all reported analyses.

References

1. Biggs R, de Vos A, Preiser R, Clements H, Maciejewski K, Schlüter M, editors. The Routledge Handbook of Research Methods for Social-Ecological Systems. London: Routledge; 2021. Available from: https://doi.org/10.4324/9781003021339.

2. Palacios O, Barceló JA, Delgado R. Exploring the role of ecology and social organisation in agropastoral societies: A Bayesian network approach. PLOS ONE. 2022;17(10):e0276088. doi:10.1371/journal.pone.0276088.

3. Watts J, Jackson JC, Arnison C, Hamerslag EM, Shaver JH, Purzycki BG. Building quantitative cross-cultural databases from ethnographic records: Promise, problems and principles. Cross-Cultural Research. 2022;56(1):62–94. doi:10.1177/10693971211065720.

4. Nimkoff MF, Middleton R. Types of family and types of economy. American Journal of Sociology. 1960;66(3):215–225. doi:10.1086/222876.

5. Pasternak B, Ember CR, Ember M. On the conditions favoring extended family households. Journal of Anthropological Research. 1976;32:109–123.

6. Netting RM. Smallholders, Householders: Farm Families and the Ecology of Intensive, Sustainable Agriculture. Stanford, CA: Stanford University Press; 1993.

7. Lévi-Strauss C. The Elementary Structures of Kinship. Revised ed. Boston: Beacon Press; 1969.

8. Wiessner P. Risk, reciprocity and social influences on !Kung San economics. In: Leacock E, Lee RB, editors. Politics and History in Band Societies. Cambridge: Cambridge University Press; 1982. p. 61–84.

9. Halstead P, O'Shea J, editors. Bad Year Economics: Cultural Responses to Risk and Uncertainty. Cambridge: Cambridge University Press; 1989.

10. Yesner DR. Maritime hunter-gatherers: ecology and prehistory. Current Anthropology. 1980;21(6):727–750. doi:10.1086/202568.

11. Makarewicz CA, Arbuckle BS, Öztan A. Vertical transhumance of sheep and goats identified by intratooth sequential carbon (δ^{13}C) and oxygen (δ^{18}O) isotopic analyses: evidence from Chalcolithic Köşk Höyük, central Turkey. Journal of Archaeological Science. 2017;86:68–80. doi:10.1016/j.jas.2017.01.003.

12. Tejedor-Rodríguez C, Moreno-García M, Tornero C, Hoffman A, García-Martínez de Lagrán Í, Arcusa-Magallón H, et al. Investigating Neolithic caprine husbandry in the Central Pyrenees: Insights from a multi-proxy study at Els Trocs cave (Bisaurri, Spain). PLOS ONE. 2021;16(1):e0244139. doi:10.1371/journal.pone.0244139.

13. Kelly RL. The Lifeways of Hunter-Gatherers: The Foraging Spectrum. Cambridge, UK: Cambridge University Press; 2013. Available from: https://doi.org/10.1017/CBO9781139176132.

14. Pearl J. Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference. San Francisco, CA: Morgan Kaufmann; 1988.

15. Hamilton MJ, Walker RS, Buchanan B. Institutional complexity emerges from socioecological complexity in small-scale human societies. Proceedings of the Royal Society B: Biological Sciences. 2024;291(2022):20240246. doi:10.1098/rspb.2024.0246.

16. D-PLACE. Database of Places, Languages, Culture and Environment; 2023. https://d-place.org/.

17. eHRAF. eHRAF World Cultures; 2023. https://hraf.yale.edu/.

18. Pearl J. Causality: Models, Reasoning, and Inference. 2nd ed. Cambridge, UK: Cambridge University Press; 2009. Available from: https://www.cambridge.org/9780521895606.

19. Spirtes P, Glymour C, Scheines R. Causation, Prediction, and Search. 2nd ed. Cambridge, MA: MIT Press; 2000. Available from: https://mitpress.mit.edu/9780262194402/.

20. Spirtes P, Glymour C. An algorithm for fast recovery of sparse causal graphs. Social Science Computer Review. 1991;9(1):62–72. doi:10.1177/089443939100900106.

21. Murdock GP. Ethnographic Atlas. Pittsburgh, PA: University of Pittsburgh Press; 1967.

22. Service ER. Primitive Social Organization: An Evolutionary Perspective. New York: Random House; 1962.

23. Peterson RA. Finding Optimal Normalizing Transformations via `bestNormalize`. The R Journal. 2021;13(1):310–329. doi:10.32614/RJ-2021-041.

24. Lauritzen SL. Graphical Models. Oxford, UK: Oxford University Press; 1996.

25. Koller D, Friedman N. Probabilistic Graphical Models: Principles and Techniques. Cambridge, MA: MIT Press; 2009. Available from: https://mitpress.mit.edu/9780262013192/.

26. Scutari M. Learning Bayesian Networks with the `bnlearn` R Package. Journal of Statistical Software. 2010;35(3):1–22. doi:10.18637/jss.v035.i03.

27. Verma T, Pearl J. Equivalence and synthesis of causal models. In: Proceedings of the Sixth Conference on Uncertainty in Artificial Intelligence (UAI-1991). Morgan Kaufmann; 1991. p. 220–227. Available from: https://dl.acm.org/doi/10.5555/1865043.1865073.

28. Colombo D, Maathuis MH. Order-independent constraint-based causal structure learning. Journal of Machine Learning Research. 2014;15:3741–3782.

29. VanderWeele TJ. Explanation in Causal Inference: Methods for Mediation and Interaction. New York: Oxford University Press; 2015.

30. Andrews RM, Foraita R, Didelez V, Witte J. A practical guide to causal discovery with cohort data; 2021. Available from: https://arxiv.org/abs/2108.13395.

31. Kalisch M, Mächler M, Colombo D, Maathuis MH, Bühlmann P. Causal Inference Using Graphical Models with the R Package `pcalg`. Journal of Statistical Software. 2012;47(11):1–26. doi:10.18637/jss.v047.i11.

32. Foraita R, Witte J. `micd`: Multiple Imputation in Causal Graph Discovery; 2025. Available from: `https://CRAN.R-project.org/package=micd`.

33. Andrews B, Ramsey J, Cooper GF. Scoring Bayesian networks of mixed variables. International Journal of Data Science and Analytics. 2018;6(1):3–18. doi:10.1007/s41060-017-0085-7.

34. Scutari M, Denis JB. Bayesian Networks: With Examples in R. 2nd ed. Boca Raton, FL: CRC Press; 2021.

35. Maathuis MH, Kalisch M, Bühlmann P. Estimating high-dimensional intervention effects from observational data. The Annals of Statistics. 2009;37(6A):3133–3164. doi:10.1214/09-AOS685.

36. Hernán MA, Robins JM. Causal Inference: What If. Boca Raton, FL: Chapman & Hall/CRC; 2020. Available from: `https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/`.

37. Hames RB, Vickers WT. Optimal diet breadth theory as a model to explain variability in Amazonian hunting. American Ethnologist. 1982;9(2):358–378. doi:10.1525/ae.1982.9.2.02a00090.

38. Ember CR, Ember M, Pasternak B. On the development of unilineal descent. Journal of Anthropological Research. 1974;30:69–94.

Figure Legends

Fig 1. Geographic distribution of the 157 small-scale farming societies analyzed. Each point represents one ethnographic society (population $< 1{,}000$) drawn from the cross-cultural dataset of Palacios et al. [2], sourced from the D-PLACE and eHRAF databases, in which agriculture and animal husbandry together constitute more than 50% of subsistence. Society locations are from D-PLACE [16]; the base map is from Natural Earth (public domain).

Fig 2. Causal Bayesian network of ecological characteristics, subsistence strategies, and social organization. Consensus network of 16 arcs (bootstrap arc strength ≥ 0.3). Dashed arcs (Gather_Agri $\rightarrow$ CommSize and CommSize $\rightarrow$ Hunt_Agri) indicate directional uncertainty, with bootstrap direction probabilities of 0.51 and 0.56, respectively. Nodes are color-coded by domain: geographical and topographical characteristics (gray), climate (yellow), subsistence strategies (green), and social organization (blue). AnnualTemp, mean annual temperature; CommOrg, community organization; CommSize, community size; DistanceCoast, distance to nearest coastline; HouseholdOrg, household organization; MonthlyPrecip, mean monthly precipitation; Hunt_Agri, Gather_Agri, Animal_Agri, and Fish_Agri, log-ratios of hunting, gathering, animal husbandry, and fishing to agriculture.

Fig 3. Interventional effects on subsistence strategies. Ratios of interventional probabilities of a high (above-median) outcome under High versus Low interventions on each continuous ancestor variable. Outcomes are Animal_Agri (top left), Hunt_Agri (top right), Gather_Agri (bottom left), and Fish_Agri (bottom right). The dashed lines mark a ratio of 1.

Fig 4. Interventional effects on social organization. Top: ratios of interventional probabilities of a large (above-median) community (left), of each household organization category (middle), and of each community organization category (right) under High versus Low interventions on continuous ancestors. Bottom: ratios of interventional probabilities of Village, Homesteads, and Hamlet settlement. The dashed lines mark a ratio of 1.

Supporting information

S1 Table. Corrections to ecological values in the compiled dataset. Entries of the five ecological variables used in the analysis that differed from, or were missing but available in, D-PLACE, for the 157 societies analyzed, with the original values in the dataset of Palacios et al. and the corrected (D-PLACE) values.

S2 Table. Bootstrap arc strengths and direction probabilities of the arcs in

the learned network (Fig 2). Arc strength is the proportion of the retained bootstrap networks containing an arc between the two variables, regardless of direction; direction probability is the proportion of these occurrences supporting the direction shown, with undirected arcs contributing equally to both directions. Arcs are listed in decreasing order of strength; all arcs with strength ≥ 0.3 are shown.

S3 Table. Bootstrap arc strengths and direction probabilities of the arcs in the discretized network (S7 Fig). All continuous variables were dichotomized at their medians, and the network was learned with the same structural constraints, significance level, and bootstrap procedure as the mixed-type network. Arc strength and direction probability are defined as in S2 Table. Arcs are listed in decreasing order of strength; all arcs with strength ≥ 0.3 are shown.

S1 Fig. Interventional effects on subsistence strategies under M1. The Gather_Agri–CommSize arc is reversed (CommSize → Gather_Agri); all other arcs are as in Fig 2. Panels as in Fig 3.

S2 Fig. Interventional effects on social organization under M1. The Gather_Agri–CommSize arc is reversed (CommSize → Gather_Agri). Panels as in Fig 4, except that the community size panel is omitted because CommSize has no ancestors in this network.

S3 Fig. Interventional effects on subsistence strategies under M2. The CommSize–Hunt_Agri arc is reversed (Hunt_Agri → CommSize); all other arcs are as in Fig 2. Panels as in Fig 3.

S4 Fig. Interventional effects on social organization under M2. The CommSize–Hunt_Agri arc is reversed (Hunt_Agri → CommSize); all other arcs are as in Fig 2. Panels as in Fig 4.

S5 Fig. Interventional effects on subsistence strategies under M3. Both the Gather_Agri–CommSize and CommSize–Hunt_Agri arcs are reversed (CommSize → Gather_Agri and Hunt_Agri → CommSize); all other arcs are as in Fig 2. Panels as in Fig 3.

S6 Fig. Interventional effects on social organization under M3. Both the Gather_Agri–CommSize and CommSize–Hunt_Agri arcs are reversed (CommSize → Gather_Agri and Hunt_Agri → CommSize); all other arcs are as in Fig 2. Panels as in Fig 4.

S7 Fig. Bayesian network learned after median-based dichotomization of all continuous variables. Consensus network of eight arcs (bootstrap arc strength ≥ 0.3). The dashed arc (CommSize → Gather_Agri) indicates directional uncertainty (direction probability 0.54). Node colors as in Fig 2.